

ANIRUDH MANTRI VEERA SAI

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PROFESSIONAL SUMMARY

AI / Python / Data Analytics professional with 4+ years of experience delivering production-grade ML systems, NLP pipelines, and analytics solutions for retail and enterprise clients. Strong hands-on expertise in Python (pandas, NumPy, scikit-learn), SQL, and large-dataset processing, with deep proficiency in GenAI, LLM orchestration, RAG, and vector search. Proven track record on AWS and GCP, automating data pipelines and presenting insights confidently to customers and executive stakeholders. Skilled in BI tools (Power BI, Tableau) and MLOps best practices.

EXPERIENCE

Glenen Labs LLC | AI Engineer

Sep 2023 – Present

Norfolk, VA | Production-Deployed Retail AI System

- Built and deployed a production-grade conversational AI kiosk for retail stores, enabling natural language product search, intelligent recommendations, and voice interaction — now live in-store with multi-location expansion underway.
- Engineered hybrid retrieval system (HNSW vector search + BM25) using pgvector and PostgreSQL, improving semantic ranking accuracy and reducing search latency from 20s to under 5s (75% improvement).
- Designed multi-agent NLP pipeline for intent detection, context tracking, and multi-turn conversation memory, achieving 100% success rate in controlled testing scenarios.
- Implemented LLM orchestration with fallback models (OpenAI GPT + Ollama) and anti-hallucination safeguards for reliable, cost-efficient inference at scale.
- Built scalable backend using FastAPI, Redis, Docker; integrated AWS Polly for emotional voice synthesis and real-time inventory data feeds.
- Developed React-based kiosk interface with voice + touchscreen interaction; optimized for zero-budget deployment using open-source tools and free-tier cloud infrastructure.

Amazon | Business Data Analyst

Mar 2021 – Aug 2023

India

- Developed Python-based ML models (scikit-learn, pandas, NumPy) for demand forecasting and customer segmentation, improving forecast accuracy by 25% and supporting data-driven decisions for business stakeholders.
- Wrote and optimized complex SQL queries on large-scale datasets (100M+ rows) to power analytics dashboards and ad-hoc analysis for executive reviews.
- Built ETL pipelines on GCP (BigQuery, Dataflow) to consolidate multi-source operational data, reducing reporting latency by 40% and enabling near real-time business insights.
- Boosted client ROI by 60% through real-time performance analytics and strategic A/B testing that optimized ad creative and targeting.
- Consolidated multi-channel data using ETL workflows, improving attribution accuracy by 40% and presenting findings directly to client stakeholders in weekly reviews.

PROJECTS

Conversational AI Assistant for Enterprise Knowledge Base

Jan 2024 – Feb 2024

- Built a multi-turn conversational AI assistant using Python, LangChain, and OpenAI GPT with RAG over 5,000+ enterprise documents stored in a vector database (FAISS), enabling natural-language Q&A with cited sources.
- Implemented intent detection, context tracking, and hybrid retrieval (semantic + keyword) to reduce hallucinations by 60%; deployed on GCP (Vertex AI, BigQuery) with MLOps practices including CI/CD, model versioning, and monitoring.

TECHNICAL SKILLS

Languages: Python (pandas, NumPy, scikit-learn), SQL, TypeScript, Java, C++, R

ML & AI: PyTorch, XGBoost, AdaBoost, FinBERT, BERT, LLM Orchestration, RAG, Vector Search (pgvector, HNSW)

Cloud & Infrastructure: AWS (EC2, S3, Polly), Google Cloud, Docker, Redis, PostgreSQL

Big Data: Apache Spark, PySpark, Kafka, Hadoop

Tools & Frameworks: FastAPI, React, Power BI, Tableau, ETL Pipelines, Data Modeling, APIs

EDUCATION

Old Dominion University — M.S., Computer Science

2023 – 2025